

From mice to humans

A multi-omic predictive framework for translational immunology — linking cross-species concordance to sequence evolution, cis-regulatory architecture and multilayer machine learning.

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github.com/wapsyed/mousetohuman_multilayer · wapsyed.github.io/mousetohuman_multilayer

0.89–
0.97

MODULE-LEVEL (BTM)
MOUSE–HUMAN
CORRELATION

0.84–
0.98

ROC-AUC · MOUSE
BTMS CALLING
HUMAN DIRECTION

R² 0.234

RANDOM-FOREST
HUMAN-RANK
REGRESSION (LOCO-
CV)

AUC
0.849

MULTILAYER
DIRECTIONAL
CONCORDANCE

01 · The problem

Murine models underpin preclinical vaccine research, yet their translational value remains **contested**. Prior cross-species comparisons were largely **descriptive** — gene-centric, linear, and disconnected from sequence evolution and cis-regulatory architecture.

201 mouse immunization datasets screened from GEO; 6 matched mouse–human conditions carried forward.

02 · Study design & harmonisation

- **Vaccination** — Fluvad® (influenza), Engerix B® (hepatitis B)
- **Infection** — *S. aureus*, *E. coli*
- **Injury** — burn, trauma-haemorrhage
- + **DMD** (mdx mouse) as a biological negative control

Blood transcriptomes · GEO / BioProject · limma + empirical Bayes · covariates (age, sex, ethnicity) · human downsampled to n = 5, 200 iterations, median statistics.

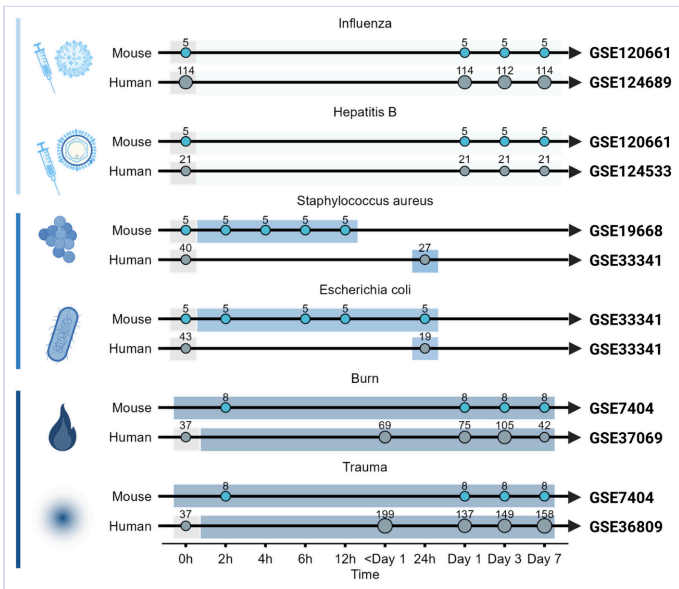


Fig. 1b — matched mouse and human timelines across vaccination, infection and injury (GSE120661 · GSE124689 · GSE19668 · GSE33341 · GSE7404 · GSE37069 · GSE36809).